WANG HOUTIAN

Undergraduate Physics Student · Nanyang Technological University

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EDUCATION

Nanyang Technological University, Singapore — **Expected 2028**

Bachelor of Science in Physics

CGPA: 4.38/5.00

Transferred from Materials Science and Engineering to Physics in 2026.

RESEARCH EXPERIENCE

From the Quantum Particle to Nonrelativistic Quantum Fields: A Mathematical Tour — **August 2026-Present**

URECA Project · Supervisor: Prof. François Gay-Balmaz · NTU

- Developing a mathematically structured route from single-particle quantum mechanics to nonrelativistic quantum field theory.
- Constructing an explicit dictionary between one-particle and second-quantized formulations, including bosonic and fermionic Fock spaces, creation and annihilation operators, quantum fields, and possible extensions to symmetry.

Memory Effects in Stochastic Thermodynamics — **May 2026-Present**

Undergraduate Research Project · Supervisor: Prof. François Gay-Balmaz · NTU

- Investigating memory effects in stochastic thermodynamics through a variational formulation proposed by Prof. François Gay-Balmaz.
- Applying the framework to the full non-Markovian system, examining fluctuation-dissipation consistency, stationary behavior, and entropy production.

RESEARCH INTERESTS

Undergraduate focus: Non-Markovian stochastic dynamics; stochastic thermodynamics; nonequilibrium statistical mechanics.
Future directions: Mathematical physics, symmetry, and representations; quantum field theory, holography, and AdS/CFT; quantum information theory.

SELECTED ACADEMIC WRITING

Analytical Mechanics — **Available on website**

Structured notes on Lagrangian and Hamiltonian mechanics.

Classical Field Theory — **In preparation**

Quantum Mechanics — **In preparation**

Mathematical Methods for Physics — **In preparation**

TECHNICAL PROJECTS AND LEADERSHIP

Tech Echo Collective — **2026-Present**

Founder and Developer · techecho.org

- Founded and developed a multilingual open-source platform for technical projects and long-form discussion, integrating GitHub authentication, project attribution, contributor histories, and GitHub Discussions.
- Created Physics Atlas, an open-source alpha platform for exploring physics research ecosystems using provenance-aware metadata from arXiv and INSPIRE.